Standardised Comparison of Convolutional and Transformer Backbones for Brain Tumour MRI Classification under Perceptual-Hash Deduplication

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Abstract

Magnetic resonance imaging (MRI) is the standard investigation for brain tumours, but manual interpretation is slow, variable between readers, and dependent on expertise that is unevenly distributed. Deep learning has been applied widely to this problem, yet reported accuracies are difficult to compare because studies differ in dataset, preprocessing, splitting and evaluation, and because near-duplicate images drawn from shared public sources frequently leak between training and test partitions. We benchmark nine ImageNet-pretrained backbones spanning classical convolutional networks (ResNet50, VGG16, DenseNet121), efficient convolutional networks (EfficientNetB0, MobileNetV3-Large, InceptionV3) and transformer or hybrid architectures (ViT-B/16, Swin-Base, CoAtNet-0) under matched data, preprocessing and schedule. Six of the nine were additionally evaluated on three-way subtype classification. Perceptual hashing applied before splitting removed 19.3% near-duplicate images from the multiclass collection, reducing 5,828 MRI images to 4,705 unique samples; binary detection used a separate set of 3,253 images, which was not deduplicated before training and which a post-hoc audit with the same procedure shows to be highly redundant: 82.6% of its test images have a near-duplicate in its training split. All models were initialised from ImageNet weights and fine-tuned using the AdamW optimiser, a cosine annealing learning rate schedule, and mixed precision training on two NVIDIA Tesla T4 graphics processing units. On the held-out test set, CoAtNet-0, DenseNet121, EfficientNetB0, and ResNet50 each reached 99.39% binary detection accuracy, with DenseNet121 and ResNet50 attaining an area under the receiver operating characteristic curve (AUC) of 1.0000; on the 85 test images without a training near-duplicate, the accuracy of these four models fell to between 96.47% and 98.82%. For multiclass classification, Swin-Base achieved the highest accuracy of 98.58% with a macro-averaged one-versus-rest AUC of 0.9993, followed by CoAtNet-0 at 98.44% and InceptionV3 at 98.30%. EfficientNetB0 offered the best accuracy-to-cost balance, reaching 99.39% detection accuracy and 97.59% subtype accuracy with only 4.0 million parameters, 0.77 GFLOPs per image, and 8.9 ms single-image latency. Gradient-weighted class activation mapping indicates that the highest-accuracy subtype model attends substantially to regions outside the brain, so accuracy alone does not establish that pathology has been learned.

Keywords: Brain tumour MRI, deep learning, perceptual hashing, shortcut learning, vision transformers

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1. Introduction

Brain tumours constitute the abnormal proliferation of cells either in the brain or the central nervous system. Whether benign or malignant, brain tumours represent serious threats because of their location in the limited skull cavity and the important functions of the brain. According to the Central Brain Tumor Registry of the United States, 453,623 primary brain and other central nervous system tumours were diagnosed in the United States between 2016 and 2020, an average of approximately 90,700 cases per year [24]. The exact prevalence in India, however, is not fully documented. Three major forms of primary brain tumours include: (i) glioma, which is formed from glial cells and accounts for 26.3% of all primary brain and other central nervous system tumours in the same registry [24]; (ii) meningioma, which develops in the meningeal layers of the brain and is usually benign in nature; and (iii) pituitary tumours, which grow in the pituitary gland and cause hormonal problems [2].

Magnetic resonance imaging (MRI) is used nowadays as the main non-invasive technique for brain tumour detection and classification. In comparison to traditional computed tomography imaging, MRI gives much better soft tissue contrast without utilising any ionising radiation. Moreover, MRI allows images to be acquired in different orientations and with different types of contrast, namely T1-weighted, T2-weighted, fluid-attenuated inversion recovery (FLAIR), and diffusion-weighted. This is extremely important for determining tumour location, volume, margins, oedema, and adjacent vital structures. At the same time, analysis of MRI studies raises a number of problems that cannot be ignored. Firstly, examination takes a lot of time, because a complete brain MRI study may include hundreds of sections, each of which must be reviewed. Secondly, inter- and intra-observer variability should be noted, as there is no guarantee that radiologists will interpret the results of an MRI study in the same way. Thirdly, limited training in neuroradiology can become an obstacle to correct image interpretation. Finally, there are not enough specialists in the rural and semi-urban regions of India because of the poor radiologist-to-population ratio.

The introduction of deep learning technology has transformed the field of medical imaging analysis in the last ten years. Models based on deep learning, especially convolutional neural networks (CNNs), have the ability to learn hierarchical representations of visual features directly from pixel inputs without the need for manual feature extraction. Thus, the use of these models enables automatic, efficient, and consistent tumour detection from brain MRI images. Convolutional neural networks have become popular among researchers working in this area, with well-performing models such as ResNet, VGG, DenseNet, Inception, EfficientNet, and MobileNet setting benchmarks on image classification tasks. More recently, researchers started exploring transformer models for computer vision problems, having been introduced initially for natural language processing. These transformer-based models include the vision transformer (ViT), the Swin Transformer, and CoAtNet. This line of research provides an alternative way of representing global information in images using the self-attention mechanism, and it might be of particular use in medical imaging, especially for understanding the relation between a tumour and the surrounding anatomical structures [20, 21, 22, 23].

Although several research works have separately proved the efficiency of deep learning architectures for detecting and classifying brain tumours, there is still an insufficient number of comparative investigations performed across several architectural families using the same experimental settings. The lack of comparative analysis poses serious challenges in choosing a suitable model, since each study employs a separate database, preprocessing procedure, data augmentation approach, and assessment technique, which makes it problematic to compare results reported by different studies. For example, it is impossible to say which performance figure is better when one work reports an accuracy of 99% on a small and homogeneous database, whereas another gives 97% accuracy on a larger and heterogeneous one.

Numerous papers make the mistake of retaining images that are duplicates or near-duplicates across the training, validation, and test splits when using publicly available datasets drawn from multiple sources. This results in a highly inaccurate assessment of performance, causing a significant increase in reported performance figures compared with the actual ability of a model to generalise. Once a model has been shown a particular image, or one that is very similar, during training, any further performance result for that image reflects memorisation rather than genuine learning. The problem of near-duplicate images is especially critical within brain tumour research, since most publicly available datasets come from common origins such as the Cheng dataset or the Figshare collection. Perceptual hashing addresses images that are duplicates of one another. It does not address a second and distinct form of correlation: adjacent slices of the same patient are visually different enough to survive any hash-based filter, yet are not statistically independent, so a split drawn at slice level can place highly correlated images on both sides of the partition even when no near-duplicate pair remains.

It is important to emphasise the clinical significance of establishing a brain tumour diagnosis correctly. The potential consequences of late diagnosis or misdiagnosis include limited treatment alternatives, poor prognosis, higher healthcare expenditure, and significantly diminished quality of life for both patients and their families. The implementation of automated systems with accuracy equal or superior to that of humans may bring about several advantages: faster diagnosis through prompt preliminary assessment; decision-making support where experienced radiologists are unavailable; uniformity of diagnosis due to the absence of inter-reader variation; and relief of radiologist workload so that experts attend primarily to challenging cases [10, 11].

This paper makes four contributions. First, it establishes a matched-data, matched-preprocessing comparison of nine ImageNet-pretrained backbones spanning three architectural families on brain tumour MRI, and reports the optimisation budget each model actually received so that architecture and training budget can be separated by the reader. Second, it applies perceptual-hash deduplication before partitioning to the multiclass track and audits the binary track with the identical procedure, quantifies the near-duplicate burden of the combined public collections by source and by label, and reports how sensitive the resulting test accuracies are to the deduplication threshold. Third, it reports the quantities that determine whether a model is deployable but are routinely omitted in this literature (per-class sensitivity and specificity, expected calibration error, agreement corrected for chance, and measured latency and throughput), and shows that models with identical accuracy differ in whether their errors are missed tumours or false alarms. Fourth, it audits whether the highest-accuracy model attends to pathology, and finds evidence that it does not, which constrains how the leaderboard should be interpreted; for the released EfficientNetB0 checkpoints it also measures the share of attention falling outside the head and the accuracy obtainable from background alone.

The rest of this paper is structured as follows. Section 2 reviews existing studies on brain tumour classification. Section 3 describes the complete methodology adopted in this study, including the overall workflow, data collection, preprocessing, model development, and experimental settings. Section 4 presents and discusses the results of both the binary detection and the multiclass classification tasks, together with efficiency and interpretability analyses. Section 5 concludes the study and offers directions for future research. Per-class tables, complete efficiency and training tables, and per-model figures are provided in the Supplementary Material, cited as Supplementary Tables S4–S12 and Figures S1–S25.

2. Literature Review

Previous works on automated brain tumour recognition were based on feature engineering followed by machine learning classifiers. Such works have used texture descriptors such as the grey level co-occurrence matrix (GLCM) and local binary patterns (LBP), wavelet transforms such as the discrete wavelet transform (DWT), morphological features related to shape, size, and boundary, and classifiers such as the support vector machine (SVM), random forests, and *k*-nearest neighbours [2, 3]. In their work, Zacharaki et al. [1] obtained around 85% accuracy using texture features and an SVM classifier on a small set of 80 images for tumour type recognition. Cheng et al. [5] suggested a tumour region augmentation approach that yielded 90% accuracy using intensity, texture, and shape features along with an SVM classifier on 3,064 MRI images. Although promising, such methods faced challenges due to the limitations of feature engineering, the inability to generalise across datasets, and the absence of automatic learning of hierarchical structures.

The success of AlexNet in 2012 propelled the implementation of deep learning in medical imaging applications. Successive CNN models improved performance by introducing better architectures. He et al. [6] incorporated residual connections into CNN models, making it possible to train very deep networks by addressing the vanishing gradient problem. ResNet50 became a standard model for medical image classification, with reports of 95–98% accuracy for brain tumour classification. Using transfer learning on a ResNet50 model, Seetha and Raja [9] achieved an accuracy of 97.5% for the classification of 253 brain MRI images. Simonyan and Zisserman [4] demonstrated improvements in model performance by increasing depth while keeping small convolutional filters of size 3×3. However, due to its very large parameter count of 138 million, VGG16 is computationally intensive and vulnerable to overfitting when applied to small medical datasets. For instance, Khan et al. [16] reported 98.69% accuracy using VGG16 with 3,064 samples. Huang et al. [8] introduced densely connected layers in which each layer receives input from all preceding layers. Consequently, DenseNet121 became more parameter-efficient while achieving better performance in medical imaging tasks, with accuracies of 96–98%. Szegedy et al. [7] presented parallel convolutions with varying kernel dimensions (1×1, 3×3, 5×5) that allowed the model to learn multi-scale features. InceptionV3 reached 97.5% accuracy in brain tumour classification, as reported by Deepak and Ameer [12]. Moreover, Rehman et al. [18] obtained a high classification rate of 99.34% accuracy via transfer learning. Tan and Le [14] introduced a method for scaling network dimensions, namely depth, width, and resolution, using neural architecture search to obtain better accuracy with far fewer parameters than comparable models. Furthermore, Howard et al. [15] created lightweight neural networks based on depth-wise separable convolutions for mobile and edge devices.

Dosovitskiy et al. [20] adapted transformer models to visual tasks by framing images as sequences of patches of 16×16 pixels. The ViT-B/16 variant performed comparably with CNNs on large-scale datasets such as ImageNet and JFT-300M but needed extensive pre-training and exhibited drawbacks on small medical datasets because of a lack of built-in inductive biases. Liu et al. [21] proposed hierarchical transformers using shifted windows to process images efficiently at multiple scales while retaining linear complexity with respect to image resolution. Swin Transformers performed impressively on ImageNet and have also been investigated for medical image processing. The hierarchical window attention in Swin Transformers enables the model to learn both local tumour boundaries and overall structure from MRI images. CoAtNet, proposed by Dai et al. [22], stacks depth-wise convolution stages and relative self-attention stages vertically within a single model, which is intended to combine the inductive bias of convolution with the global receptive field of attention.

Since 2022 these architectures have been applied to brain tumour MRI directly, and the reliability of the public benchmarks on which they are evaluated has itself come under scrutiny. Shamshad et al. [27] survey the adoption of transformers across medical imaging tasks, including classification. Tummala et al. [25] fine-tuned an ensemble of four ImageNet-pretrained ViT variants on the 3,064-image Cheng collection and reported 98.7% test accuracy for three-way subtype classification, and Asiri et al. [26] reported 97% accuracy for a Swin Transformer on a 2,870-image four-class Kaggle collection. Over the same period, Kapoor and Narayanan [28] documented data leakage, including leakage between training and test data, as a recurring cause of over-optimistic results in machine-learning-based science across 17 fields. Most directly relevant here, Vangala et al. [29] audited public brain tumour MRI classification benchmarks and found that 28.8% of the official test images of one widely used collection have a near-duplicate in its training split, and that 22.3% of the test images of the Bhuvaji collection used in the present study are byte-identical to a training image. These findings strengthen the case for deduplicating before splitting rather than relying on the official partitions.

Studies applying these architectures to brain tumour detection and classification are summarised in Table 1.

Table 1: Comparison of existing studies for brain tumour detection.

Study	Model	Dataset size	Task	Accuracy
Talo et al. (2019) [13]	ResNet50	253	Binary	95.23%
Khan et al. (2020) [16]	VGG16	3,064	Binary	98.69%
Noreen et al. (2020) [17]	DenseNet121	3,064	Binary	99.51%
Rehman et al. (2020) [18]	InceptionV3	3,064	Binary	99.34%
Ghassemi et al. (2020) [19]	EfficientNet	3,064	Binary	98.87%
Cheng et al. (2015) [5]	SVM + features	3,064	Multiclass	90.00%
Deepak and Ameer (2019) [12]	InceptionV3	3,064	Multiclass	97.50%
Tummala et al. (2022) [25]	ViT ensemble	3,064	Multiclass	98.70%
Asiri et al. (2024) [26]	Swin Transformer	2,870	Multiclass	97.00%

Several open issues remain. Cross-architecture comparison is difficult because most studies examine no more than two or three models. Dataset heterogeneity is a further obstacle, because split schemes, preprocessing, and evaluation protocols differ between studies. Transformer under-representation is an additional problem, since current papers deal mainly with CNN-based models. Reproducibility is also limited, because source code, exact splits, and pretrained weights are not consistently released. Incomplete deduplication is a final gap, as most papers do not account for duplicate and near-duplicate images in combined datasets. Reporting practice is also narrow: headline accuracy and AUC are commonly quoted without per-class sensitivity and specificity, without any assessment of probability calibration, and without measured inference cost, all of which matter for clinical interpretation. This paper addresses these gaps by evaluating nine architectures under identical conditions, combining datasets from multiple public sources, applying perceptual hash based deduplication before splitting, comparing CNN and transformer families directly, reporting per-class and calibration metrics alongside measured latency and throughput, and releasing the complete code and per-model artefacts for both the binary and the multiclass tasks.

3. Methods and Materials

3.1. Overall Methodology

The complete research workflow adopted in this study is illustrated in Figure 1. The pipeline proceeds through five stages. In the data acquisition stage, two publicly available Kaggle collections are combined for the binary detection task and two further collections are combined for the multiclass subtype task. In the data curation stage, perceptual hashing is applied to the multiclass collection to remove near-duplicate images before any split is drawn, after which both datasets are partitioned into training, validation, and test subsets using stratified sampling so that class proportions are preserved in every subset. In the preprocessing stage, images are converted to three-channel form, resized to the input resolution expected by each backbone, normalised using ImageNet statistics, and augmented on the fly during training only. In the model development stage, nine ImageNet-pretrained backbones are fine-tuned under a single shared training protocol, with the classification head of each backbone replaced to match the number of target classes. In the evaluation stage, the checkpoint selected on the validation split is applied once to the held-out test split, and accuracy, per-class sensitivity and specificity, F1 score, AUC, calibration error, gradient-weighted class activation maps, and inference cost are recorded for every model.

A design point worth stating explicitly is that the two tasks are kept separate rather than merged into a single four-class problem. The binary detector answers whether a tumour-like abnormality is present, whereas subtype classification is only meaningful once a tumour-positive decision has already been taken. Keeping the tasks separate mirrors the clinical order of interpretation and avoids making the subtype model responsible for rejecting tumour-negative images that were never part of its training objective.

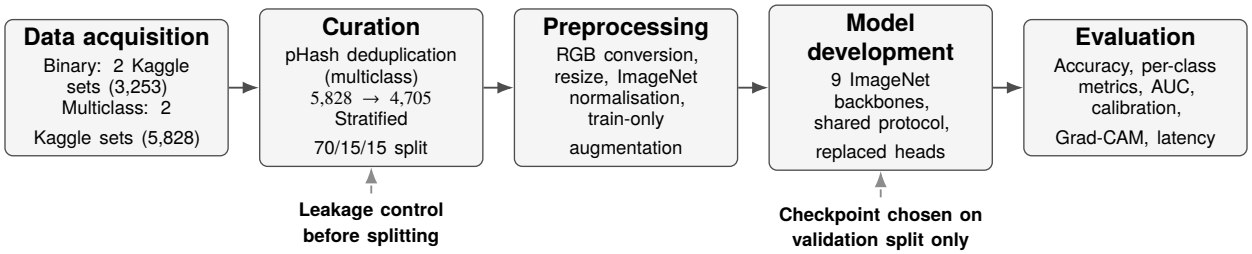


Figure 1: Overall methodology of the study, showing the five stages of the pipeline from data acquisition through curation, preprocessing, and model development to evaluation. In the multiclass track, perceptual hash deduplication is applied before splitting so that near-duplicate images cannot leak between training and test subsets; the binary track was not deduplicated and is audited for leakage in Section 3.3. The test split is used only once, with the checkpoint selected on the validation split.

3.2. Data Collection

For the binary brain tumour detection task, which requires classifying MRI images as either tumour-positive or tumour-negative, we combined two publicly available brain MRI datasets sourced from the Kaggle platform. Dataset 1, *Brain MRI Images for Brain Tumor Detection* [30], contains 253 MRI images in total, with 155 labelled as tumour-positive and 98 as tumour-negative. Dataset 2, *Br35H: Brain Tumor Detection 2020* [31], contains a substantially larger collection of 3,000 MRI images with a perfectly balanced distribution of 1,500 tumour-positive and 1,500 tumour-negative images. The images are provided in JPEG format and have been pre-cropped to focus on the brain region, though some variability in field of view and image resolution exists across samples. The combined dataset for binary classification comprised 3,253 images, with 1,598 tumour-negative images (49.1%) and 1,655 tumour-positive images (50.9%). This near-perfect balance prevents the model from exploiting class imbalance and keeps chance-level accuracy at approximately 50%. The negative class consists of images that the dataset curators labelled as not containing a tumour; neither source certifies that these images are radiologically normal, so the absence of other pathology is not established, and the class is accordingly referred to as tumour-negative rather than normal throughout. Representative samples of tumour-negative and tumour-positive images are shown in Figure 2, and the class distribution is given in Supplementary Figure S3.

For the more challenging multiclass classification task, in which the model must distinguish between glioma, meningioma, and pituitary tumours, we combined two further publicly available brain tumour MRI datasets. Dataset 1, Bhuvaji et al., *Brain Tumor Classification (MRI)* [32], provides JPEG images at variable native resolution; its training and testing folders were merged, and its tumour-negative folder was excluded because the multiclass task contains tumour classes only. Dataset 2 is the brain tumour collection of Cheng et al. [33], described in the associated study [5], which comprises 3,064 contrast-enhanced T1-weighted slices from 233 patients, and was obtained in the JPEG conversion distributed on Kaggle [34], in which 99.5% of the slices are 512×512 pixels and the remainder 256×256. The Bhuvaji collection is widely understood to be derived in substantial part from the Cheng data; at the deduplication tolerance used here, however, only 3 Cheng images were matched to a Bhuvaji image (Section 3.3). The combined dataset prior to deduplication consisted of 5,828 images, with glioma the largest subset at 2,352 images (40.4%), meningioma at 1,645 images (28.2%), and pituitary at 1,831 images (31.4%). Representative samples of each tumour type are shown in Figure 3.

The provenance of every image is summarised in Table 2, which gives for each source the number of images contributed, the number surviving deduplication, the class-by-source contingency, the native resolution, and the recorded MRI sequence. The Bhuvaji collection is listed by its original training and testing folders, which were merged before splitting. Because the original deduplication pass recorded only per-class totals, the per-source figures were obtained by reproducing that pass exactly, with the same folder order, hash, and tolerance; the reproduction retains 2,000, 1,304, and 1,401 images per class, matching the original run, and the stratified splits drawn from it with seed 42 match the stored test and validation predictions of all 15 trained models. The per-class survival of each source is given in Supplementary Table S11, and the native image sizes of each source are plotted in Supplementary Figure S17.

The MRI sequence is unknown in both experimental tracks. Neither binary source records the sequence (T1-weighted, contrast-enhanced T1-weighted, T2-weighted, or FLAIR) of individual images. In the multiclass track, the Cheng component is contrast-enhanced T1-weighted according to its source publication [5], but the Bhuvaji collection records no sequence information, so the pooled multiclass set is a mixture of unknown sequence composition. This limits clinical interpretation: a model trained on an unknown mixture of sequences cannot be specified for deployment on any particular acquisition protocol.

Table 2: Provenance of the source collections, giving the images contributed by each source, the images surviving perceptual-hash deduplication (Kept), the class-by-source contingency before deduplication, the native resolution, and the recorded MRI sequence. Native resolution is given as the number of distinct image sizes and the most common (modal) size, with the share of images at that size. Per-class survival of each source is given in Supplementary Table S11.

Track	Source	Images	Kept	Class counts			Sizes	Modal size	Sequence
				Tumour	Tumour-negative				
Binary	Chakrabarty [30]	253	206[a]	155	98		184	225 × 225 (7.1%)	Not recorded
	Br35H [31]	3,000	981[a]	1,500	1,500		291	225 × 225 (10.9%)	Not recorded
	Total	**3,253**	**1,187**[a]	**1,655**	**1,598**				
				Glioma	Meningioma	Pituitary			
Multiclass	Bhuvaji [32], training	2,475	1,925	826	822	827	137	512 × 512 (93.1%)	Not recorded
	Bhuvaji [32], testing	289	212	100	115	74	184	512 × 512 (11.1%)	Not recorded
	Cheng [33, 34]	3,064	2,568	1,426	708	930	2	512 × 512 (99.5%)	Contrast-enhanced T1
	Total	**5,828**	**4,705**	**2,352**	**1,645**	**1,831**			

[a]Not deduplicated before training; images the same pass retains when applied post hoc (Section 3.3).

The two collections differ substantially in their native image geometry, as summarised in Table 3 and illustrated in Supplementary Figures S1 and S2. The binary detection images are heterogeneous, with widths ranging from 150 to 1,920 pixels and a standard deviation of 155 to 195 pixels about mean dimensions of roughly 300 to 400 pixels, reflecting the varied cropping and field of view of the two contributing sources. The multiclass images are far more uniform, with a median of 512 × 512 pixels in every class and interquartile ranges that collapse to a single value, since the great majority of images were distributed at that resolution. The uniformity is not shared by every multiclass source, however: 99.5% of the Cheng slices and 93.1% of the Bhuvaji training folder are 512×512 pixels, against only 11.1% of the Bhuvaji testing folder, whose 289 images span 184 distinct sizes (Table 2). The difference between the two tracks matters for interpretation: the multiclass task operates on a visually homogeneous collection, whereas the binary task requires the model to absorb considerably more variation in scale and framing. All images were resized to a common input resolution during preprocessing, so this heterogeneity affects the effective magnification of anatomy rather than the tensor shape presented to the network.

Table 3: Native image dimensions of the source collections before resizing, in pixels. Values are computed across all images of each class prior to deduplication.

Task	Class	Images	Mean W	Mean H	SD W	Min W	Max W
Binary	No tumour	1,598	302.6	308.9	154.9	150	1,920
	Tumour	1,655	351.5	400.2	195.2	175	1,275
Multiclass	Glioma	2,352	507.6	508.4	33.0	175	640
	Meningioma	1,645	483.1	485.9	90.4	174	1,275
	Pituitary	1,831	512.5	512.2	84.4	201	1,365

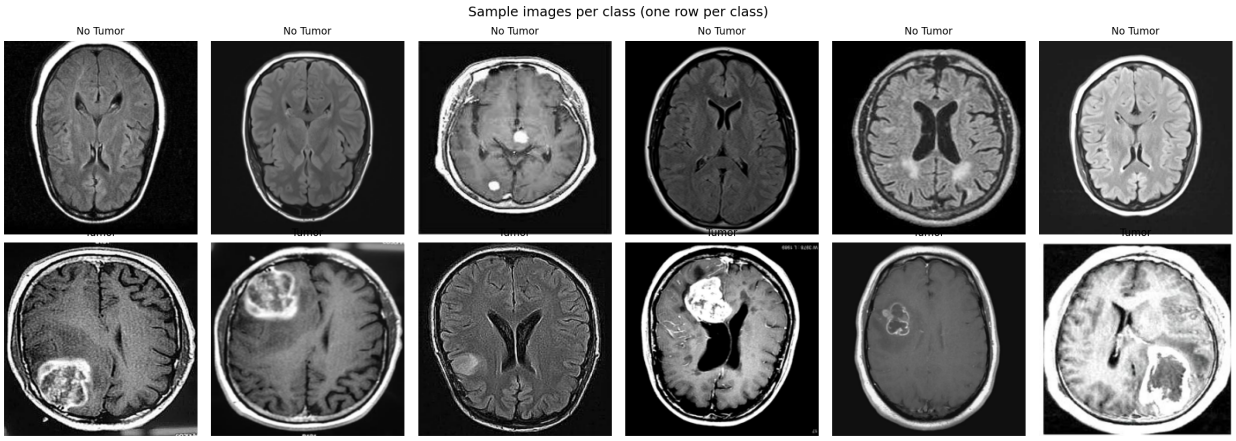


Figure 2: Sample brain MRI images used for the binary detection task. The top row shows tumour-negative images and the bottom row shows tumour-positive images.

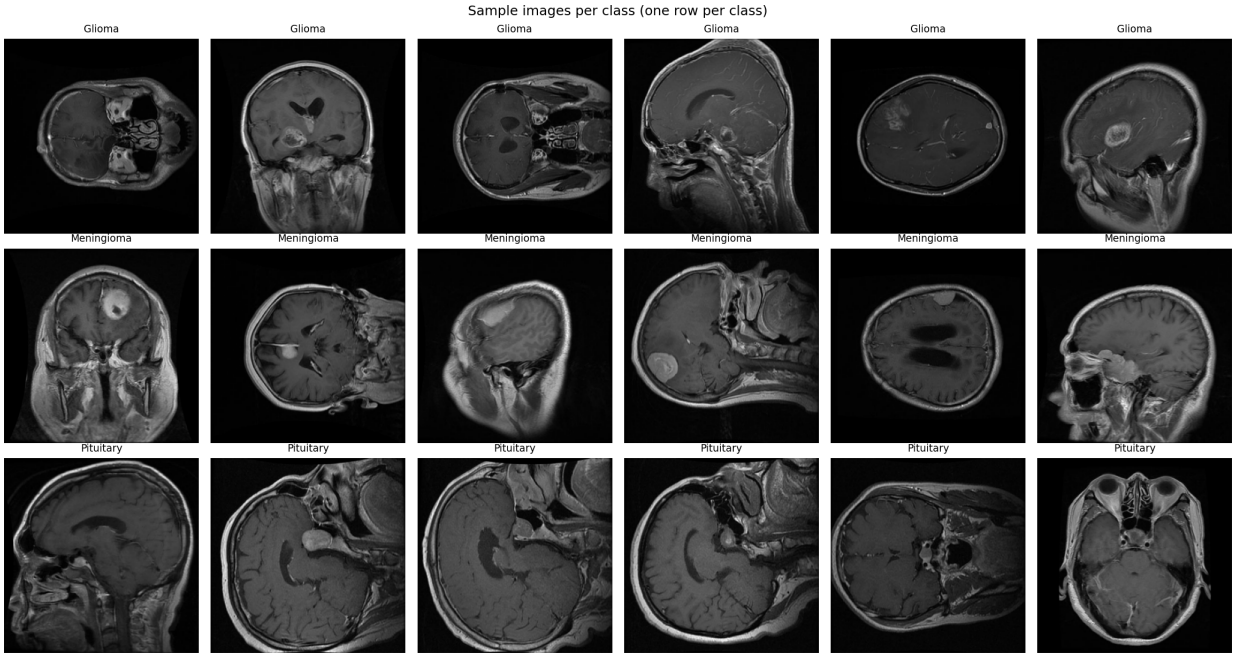


Figure 3: Sample MRI images by tumour type for the multiclass classification task, showing glioma, meningioma, and pituitary cases.

3.3. Preprocessing and Deduplication

When working with multiple datasets there is always a possibility of encountering duplicates, or images of the same underlying content that differ only in resizing or compression. These near-duplicates, if inadvertently distributed across training and test splits, constitute a form of data leakage that artificially inflates reported performance. To address this, we employed perceptual hashing (pHash), which generates a compact fingerprint for each image that is robust to resizing, compression, slight rotation, and minor contrast adjustment. Specifically, 64-bit perceptual hashes were generated using the ImageHash library. The Hamming distance between two hash values served as the similarity metric. We adopted a tolerance threshold of 5, so that any pair of images differing in at most 5 of the 64 bits was treated as a near-duplicate pair, and all but one representative of each such group were removed. The rationale for rigorous deduplication is that multiple public datasets often contain the same underlying images because they derive from common original sources, that near-duplicates cause leakage if present across train and test splits, and that removal of duplicates ensures reported metrics reflect genuine generalisation.

The adequacy of the tolerance was checked after training against an independent detector that treats two images as the same when their 64×64 greyscale thumbnails have a normalised cross-correlation (NCC) of at least 0.98, following Vangala et al. [29]. At a tolerance of 5, the hash recovers 96.9% of the 385 NCC duplicate pairs in the multiclass pool and 89.0% of the 6,078 in the binary pool, while also flagging 1,250 and 986 pairs that the NCC criterion does not count as duplicates (Supplementary Figure S19). The tolerance therefore misses a small share of near-identical pairs and removes a larger number of similar but not identical images. No pair in either pool lies at an odd Hamming distance of 11 or less, so tolerances of 4 and 5 give identical results; at tolerances of 1, 3, 5, 7, and 9 the multiclass pass would retain 5,571, 5,324, 4,705, 3,809, and 2,874 images.

Deduplication was applied to the multiclass collection, where the two constituent datasets were expected to overlap substantially. After applying perceptual hashing, the glioma class decreased from 2,352 to 2,000 images, a reduction of 14.97%; meningioma decreased from 1,645 to 1,304 images, a reduction of 20.73%; and pituitary decreased from 1,831 to 1,401 images, a reduction of 23.48%. Overall, the multiclass dataset was reduced from 5,828 to 4,705 images, a reduction of 19.3%. The class distribution before and after deduplication is shown in Figure 4. Contrary to the expectation of cross-source overlap, the redundancy removed was almost entirely within-source. Of the 1,123 removed images, only 3 were Cheng slices whose retained near-duplicate came from the Bhuvaji collection; 493 were removed against another Cheng slice and 627 against another Bhuvaji image, including 150 images of the Bhuvaji training folder whose retained near-duplicate lies in its testing folder, an overlap between that collection's official partitions consistent with the finding of Vangala et al. [29]. By construction no multiclass test image lies within Hamming distance 5 of a training image; the NCC detector finds 3 test images (0.4%) with a near-identical training image at distances of 6 and 8 that the hash did not catch.

It should be stated explicitly that perceptual hash deduplication was *not* applied to the binary detection dataset before training, on the assumption that its two source collections were disjoint in origin. Running the identical pass on the 3,253 binary images after training shows that assumption to be wrong and the asymmetry between the tracks to be material. The pass retains only 1,187 images, 459 tumour-negative and 728 tumour-positive, removing 2,066 (63.5%). Of the removals, 731 are Br35H images whose retained near-duplicate is a Chakrabarty image, so the larger collection contains near-copies of the smaller one, and the remaining 1,335 are within-source, 1,288 in Br35H and 47 in Chakrabarty; no near-duplicate pair crosses the two labels. Because the binary splits were drawn from the undeduplicated pool, 403 of the 488 binary test images (82.6%) have a training image within Hamming distance 5, and 396 (81.1%) have one meeting the NCC criterion (Supplementary Figure S20). The binary results in Section 4.1 are therefore inflated by train-test leakage. The nine binary models were not retrained on the deduplicated pool; as a proxy, Section 4.1 reports their accuracy on the 85 test images that have no training near-duplicate.

Stratified sampling was used for both splits so that every subset retained a representative distribution of the classes. For binary detection, the 3,253 images were divided into training, validation, and test subsets in the proportions 70%, 15%, and 15%, giving 2,277, 488, and 488 images respectively. For multiclass classification, the same proportions were applied to the 4,705 deduplicated images, giving 3,293 training, 706 validation, and 706 test images. The resulting per-class split sizes are reported in Table 4.

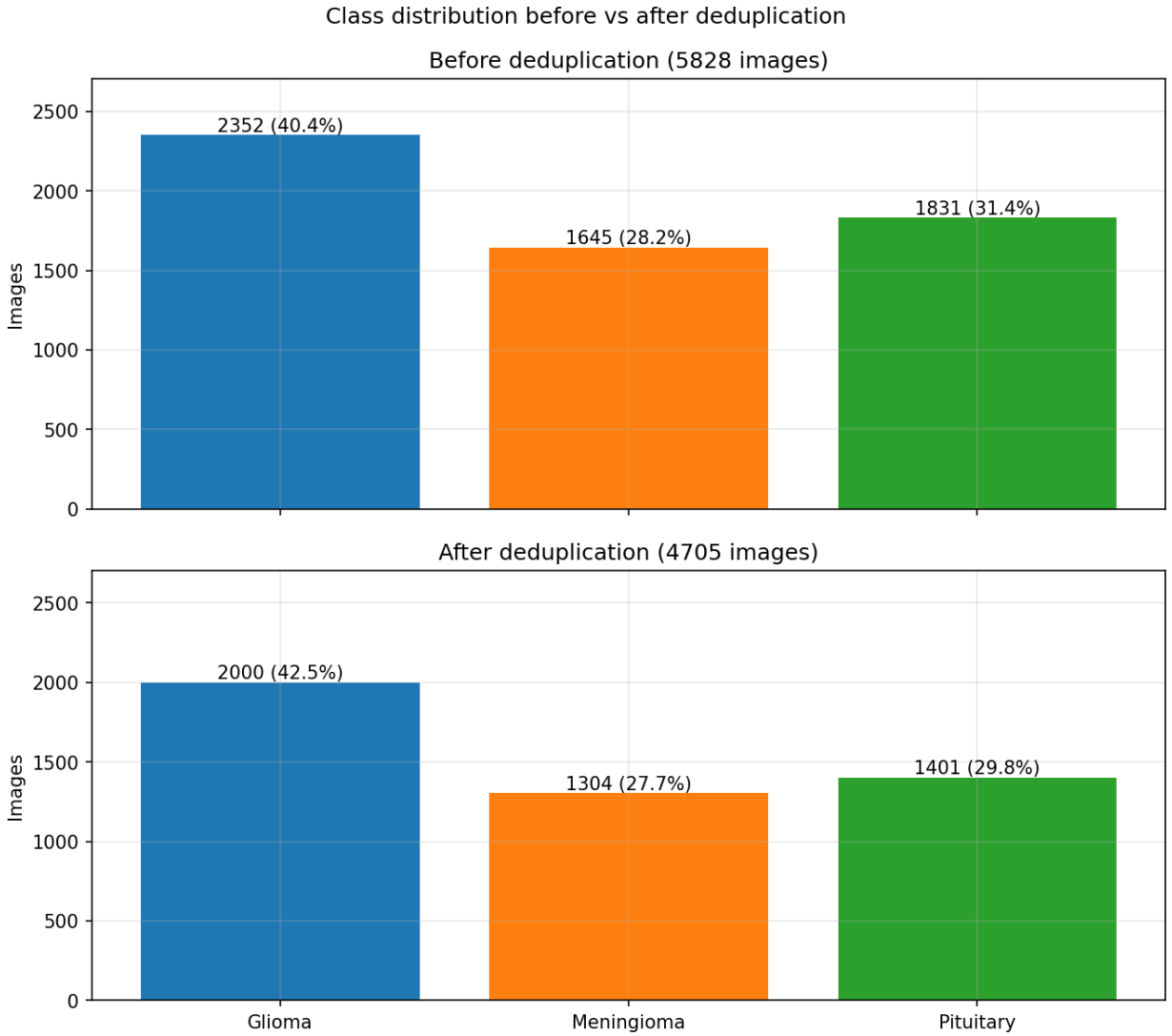


Figure 4: Class distribution of the multiclass dataset before and after perceptual hash deduplication. The total decreased from 5,828 to 4,705 images, a reduction of 19.3%.

Table 4: Stratified dataset splits for both tasks. Proportions are 70% training, 15% validation, and 15% test in each case.

Task	Class	Training	Validation	Test
Binary detection	No tumour	1,119	239	240
	Tumour	1,158	249	248
	Total	**2,277**	**488**	**488**
Multiclass	Glioma	1,400	300	300
	Meningioma	913	195	196
	Pituitary	980	211	210
	Total	**3,293**	**706**	**706**

All images were loaded using the Python Imaging Library and converted to the RGB colour format even where the original was greyscale, ensuring compatibility with the ImageNet-pretrained backbones. Images were resized to the input resolution expected by each backbone, which was 224×224 pixels for every model except InceptionV3, which requires 299×299 pixels. Pixel values were converted to a PyTorch tensor and scaled to the range $[0, 1]$, then normalised using the ImageNet channel means of $[0.485, 0.456, 0.406]$ and standard deviations of $[0.229, 0.224, 0.225]$. To improve generalisation, data augmentation was applied exclusively to the training set and consisted of random horizontal flipping with probability 0.5, random rotation of up to ± 10 degrees, and colour jitter adjusting brightness and contrast by up to 20%. These augmentations were applied on the fly at each epoch. For validation and testing, only resizing, tensor conversion, and normalisation were applied, with no stochastic augmentation, ensuring deterministic and reproducible evaluation. Representative images before and after augmentation for both tasks are shown in Supplementary Figure S6.

3.4. Model Development

Nine architectures spanning three families were evaluated. All were initialised from ImageNet-pretrained weights, and in every case the final classification layer was replaced with a linear layer producing one output for the binary task or three outputs for the multiclass task. The parameter counts quoted below and used throughout the paper are the counts measured directly from the instantiated models after head replacement, and therefore differ slightly from the nominal counts commonly quoted for the ImageNet-1000 variants of the same backbones. These measured values are reported consistently in Tables 7, 8, and 9.

ResNet50 is a 50-layer residual network built around bottleneck residual blocks with skip connections that address the vanishing gradient problem, measured at 23.51 million parameters. VGG16 is a 16-layer network using only 3×3 convolutional filters stacked in increasing depth, measured at 134.26 million parameters. DenseNet121 is a 121-layer densely connected network that concatenates the feature maps of all preceding layers, maximising information flow at only 6.95 million parameters. EfficientNetB0, developed through neural architecture search and compound scaling, achieves strong accuracy with only 4.01 million parameters and 0.77 GFLOPs per image. MobileNetV3-Large uses depth-wise separable convolutions for mobile and embedded applications, measured at 4.20 million parameters and the lowest computational cost in the study at 0.43 GFLOPs per image. InceptionV3 is a 48-layer network with parallel convolutional paths of different kernel sizes within inception modules, capturing multi-scale features at 24.35 million parameters and operating at the higher input resolution of 299×299 pixels. ViT-B/16 divides the input image into 16×16 patches and processes them as a sequence with a transformer encoder, measured at 85.80 million parameters and lacking the convolutional inductive bias of the CNN backbones. Swin-Base introduces hierarchical transformers with shifted windows, achieving linear complexity and multi-scale processing at 86.74 million parameters. CoAtNet-0 combines convolutional and attention mechanisms in a unified architecture and is measured at 26.67 million parameters. We note that CoAtNet-0 is frequently described in the literature as a model of approximately 10 million parameters; the implementation used here is the `coatnet_0_rw_224` variant, whose measured size is 26.67 million parameters, and that measured value is what is reported throughout.

The multiclass track was run for six of the nine architectures, namely CoAtNet-0, EfficientNetB0, InceptionV3, ResNet50, Swin-Base, and ViT-B/16. VGG16, DenseNet121, and MobileNetV3-Large were evaluated on the binary detection task only.

3.5. Training Configuration

All experiments were conducted on dual NVIDIA Tesla T4 graphics processing units with 16 GB of GDDR6 memory each, Intel Xeon processors with four virtual central processing unit cores, and 29 GB of system memory, running Python 3.12.13 with PyTorch 2.10.0 and CUDA 12.8. To ensure fair comparison, the hyperparameters listed in Table 5 were standardised across all models. Every model was trained for 10 epochs using the AdamW optimiser, which is the Adam optimiser with decoupled weight decay regularisation, at an initial learning rate of 3×10^{-4} and the library default weight decay coefficient of 0.01, which was left unchanged rather than tuned. The learning rate was annealed using the cosine annealing learning rate scheduler with the maximum number of iterations parameter set to 10, matching the number of epochs, so that the learning rate followed a single cosine decay from its initial value to approximately zero over the course of training. The loss function was binary cross-entropy with logits loss for the binary detection task, which combines a sigmoid activation with the binary cross-entropy criterion in a numerically stable form, and cross-entropy loss for the multiclass task. Mixed precision training was used throughout, with automatic mixed precision autocasting and gradient scaling to preserve numerical stability in half precision.

Batch sizes were model-specific and set to the largest value that fitted within the available graphics memory, since a single batch size would have been infeasible for the largest transformer backbones. The batch sizes used are listed per model in Supplementary Tables S9 and S10. This has an important consequence that must be stated plainly: because the learning rate and epoch count are held fixed while the batch size varies from 32 to 256, the number of optimiser steps each model receives varies correspondingly, from roughly 80 to 350 over the ten epochs. The comparison is therefore controlled for data, schedule, and preprocessing, but it is *not* controlled for optimisation budget, and a model trained at batch size 256 receives far fewer gradient updates than one trained at batch size 64. Differences between models consequently confound architecture with optimisation budget to an extent that cannot be separated from these runs alone.

The training loader used `drop_last=True`, so any final partial batch was discarded at each epoch. At batch size 256 this means 2,048 of the 2,277 binary training images were used per epoch, with 229 images dropped; because the loader reshuffles, the dropped subset differs from epoch to epoch, so all images are seen over the course of training but no single epoch sees the full set. The training loop at each epoch consisted of a training phase with augmentation and weight updates followed by a validation phase without augmentation. Learning rate scheduling was applied after each epoch. Model checkpointing retained the weights giving the best validation AUC for the binary task and the best validation macro AUC for the multiclass task, with the first epoch attaining the maximum retained.

The random seed was fixed at 42 for PyTorch, NumPy, and the Python standard library random module. Reproducibility is nevertheless only approximate rather than exact, because `cudnn.benchmark` was enabled, TF32 matrix multiplication was permitted, and automatic mixed precision was used throughout; each of these admits nondeterministic kernel selection. Re-running a notebook should reproduce the reported figures closely but not necessarily bit-for-bit.

Table 5: Training hyperparameters, standardised across all models and both tasks.

Hyperparameter	Setting
Epochs	10
Optimiser	AdamW (Adam with decoupled weight decay)
Weight decay	0.01
Initial learning rate	3×10^{-4}
Learning rate scheduler	Cosine annealing, maximum iterations = 10
Loss (binary detection)	Binary cross-entropy with logits loss
Loss (multiclass)	Cross-entropy loss
Precision	Mixed precision with automatic casting and gradient scaling
Batch size	Model-specific; largest fitting in memory (see Supplementary Tables S9 and S10)
Input resolution	224×224 pixels; 299×299 pixels for InceptionV3
Checkpoint selection	Best validation AUC (binary), best validation macro one-versus-rest AUC (multiclass); first epoch attaining the maximum
Random seed	42

3.6. Implementation Differences Between Models

Each architecture was implemented in its own notebook from a shared template, and the intended difference between notebooks is the backbone alone. In practice a small number of further differences were necessary or arose incidentally, and since several of them bear on the interpretation of the results they are documented here rather than left implicit. They are summarised in Table 6.

Two differences are unavoidable. First, each architecture exposes its classification head under a different attribute, so the head replacement code necessarily differs. Second, the layer targeted for gradient-weighted class activation mapping must be the last spatial or token stage of each architecture, which differs by design; for VGG16 this resolves to a max-pooling rather than a convolutional layer, which makes its attention maps less directly comparable with the others.

One difference is an inconsistency that should be taken into account when reading the two tasks together. Most backbones were obtained from `torchvision`; CoAtNet-0 was obtained from `timm` in both tasks, since CoAtNet has no `torchvision` implementation. Swin-Base, however, was not sourced consistently: the binary detection notebook uses the `timm` model `swin_base_patch4_window7_224`, whereas the multiclass notebook uses the `torchvision` model `swin_b`. These are different implementations of the Swin-B architecture and they load different pretrained checkpoints, so the two rows labelled Swin-Base in Tables 7 and 8 are not the same model initialised the same way. Their measured parameter counts differ only marginally, at 86.74 and 86.75 million, but the comparison of Swin-Base across the two tasks is not strictly like-for-like, and the contrast between its weak binary result of 95.90% and its leading multiclass result of 98.58% should not be attributed to task difficulty alone.

Two differences affect results and deserve particular attention. The InceptionV3 binary notebook applies the standard auxiliary-classifier recipe, adding 0.4 times the auxiliary loss to the main loss during training, which no other model receives. Separately, although the InceptionV3 binary model is wrapped for two-GPU data parallelism, the training loop calls the unwrapped module, so the run executed on a single GPU; the recorded peak memory of 0 MB on the second device confirms this. Its training time in Supplementary Table S9 is therefore a single-GPU measurement and is not comparable with the other rows, a point returned to in Section 4.3. In the multiclass InceptionV3 notebook the auxiliary head is disabled but not removed, so 769,000 parameters belonging to an unused 1000-class auxiliary classifier remain in the model and are counted in its reported total of 25.12 million. This accounts for the difference between the binary and multiclass InceptionV3 parameter counts in Tables 7 and 8, and it is the reason those two figures are not identical despite the backbone being the same.

Table 6: Implementation differences between the per-model notebooks, with the reason for each. Differences confined to comments, cell titles, or equivalent reformatting are not listed.

Difference	Scope	Reason
Head replacement code	All models	Each architecture exposes its classifier under a different attribute; unavoidable.
Source library `timm`	CoAtNet-0 (both tasks)	CoAtNet has no `torchvision` implementation.
Source library differs *between tasks*	Swin-Base	`timm` `swin_base_patch4_window7_224` for binary, `torchvision` `swin_b` for multiclass. Different implementations and different pretrained checkpoints; no stated reason.
Input resolution 299×299	InceptionV3	The architecture's native input size.
Batch size (32–256)	All models	Set to the largest value fitting in 16 GB of graphics memory. Only InceptionV3 states this in code; for the others it is inferred from recorded peak memory. Changes the optimisation budget.
Auxiliary loss 0.4×	InceptionV3 (binary)	Standard Inception training recipe; applied without comment.
Single-GPU execution	InceptionV3 (binary)	The training loop calls the unwrapped module, bypassing data parallelism. Unintended.
Retained unused auxiliary head	InceptionV3 (multiclass)	Only the main head was replaced; adds 769k parameters to the reported total. Unintended.
Grad-CAM target layer	All models	Must be the final spatial or token stage of each architecture. For VGG16 this is a max-pooling layer.
Pinned `IMAGENET1K_V1` weights	MobileNetV3-Large	The only model not using the library default weights. No stated reason.

A further asymmetry concerns the deduplication procedure itself. The perceptual hash pass operates over the combined multiclass pool without regard to class label, and processes classes in a fixed order, so where a near-duplicate pair spans two classes the glioma image is always the one retained. The procedure itself does not count such cross-label removals, so they were counted afterwards on the reproduced pass. Within the tolerance, 31 near-duplicate pairs cross two labels, all of them glioma against meningioma, and 23 of the 1,123 removals were made against an image of another class. Each cross-label pair was classified by its thumbnail NCC (Supplementary Figure S18). Eight pairs, involving 10 distinct images, are near-identical images filed under both labels (NCC ≥ 0.98), all within the Bhuvaji collection, which is label noise in that source; the original pass removed the 8 meningioma copies and kept 2 glioma images in the training split, and none reached the test split. The other 23 pairs are visually distinct images (NCC < 0.98), mostly Cheng slices, and are hash false positives at this tolerance. A label-aware pass that excludes both copies of every near-identical cross-label pair and deduplicates within each class only would retain 1,998 glioma, 1,310 meningioma, and 1,401 pituitary images, against 2,000, 1,304, and 1,401 in the original pass. Order dependence therefore affects only a handful of images, and the class-wise reduction percentages in Section 3.3 reflect redundancy within each class; the models were not retrained on the label-aware pool.

Finally, the multiclass source collection was assembled by merging the official training and testing folders of the Bhuvaji dataset into a single pool before applying deduplication and drawing new stratified splits. This was done so that both tasks would use the same 70/15/15 splitting procedure, but it means the multiclass results are not directly comparable with published figures that evaluate on that dataset's original test split, a point returned to in Section 4.5.

4. Results and Discussion

This section presents the findings of the binary detection and multiclass classification experiments. Section 4.1 reports binary tumour detection, Section 4.2 reports classification into glioma, meningioma, and pituitary subtypes, Section 4.3 compares computational efficiency, Section 4.4 examines model interpretability through gradient-weighted class activation mapping, and Section 4.5 compares the present results with previously published work. Unless stated otherwise, every figure quoted is measured on the held-out test split.

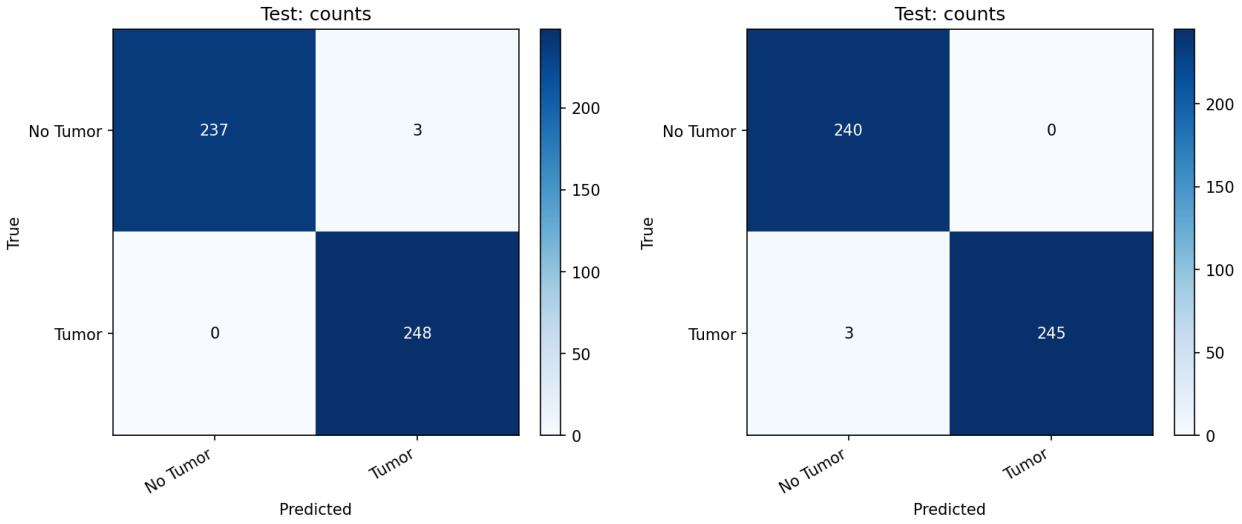
4.1. Binary Detection Results

Test-split results for the binary detection task are reported in Table 7, and the per-class breakdown for every model is given in Supplementary Table S4. Four architectures tied at the highest accuracy of 99.39%, corresponding to three misclassified images out of 488: CoAtNet-0, DenseNet121, EfficientNetB0, and ResNet50. Although the accuracy of these four models is identical, the structure of their errors differs in a way that matters clinically. CoAtNet-0 and EfficientNetB0 achieved perfect sensitivity of 100.00% at a specificity of 98.75%, so every tumour-positive test image was detected and the three errors were false alarms on tumour-negative images, whereas DenseNet121 and ResNet50 showed the mirror-image behaviour, achieving perfect specificity of 100.00% at a sensitivity of 98.79%, so their three errors were missed tumours. Figure 5 shows this contrast directly for EfficientNetB0 and ResNet50. For a screening application in which a missed tumour is the more serious failure, the error profile of CoAtNet-0 and EfficientNetB0 is preferable despite the identical headline accuracy, although DenseNet121 and ResNet50 produced the strongest threshold-independent ranking, both reaching an AUC of 1.0000 to four decimal places. The four leaders should nevertheless be regarded as statistically indistinguishable rather than genuinely tied: on a test set of 488 images a single misclassification is worth 0.20 percentage points of accuracy, each model was trained once from a single seed, no confidence intervals are available, and the separation between the four leaders and InceptionV3 amounts to three images. The top of Table 7 therefore identifies a group of models that perform equivalently on this data rather than ordering them, whereas the distinction between that group and the weaker models spans tens of images and is not in doubt. All of these figures are, however, inflated by the train-test leakage described in Section 3.3. On the 85 test images without a training near-duplicate, 62 tumour-positive and 23 tumour-negative, accuracy falls to 98.82% for CoAtNet-0, 97.65% for DenseNet121, and 96.47% for EfficientNetB0 and ResNet50 (final column of Table 7); all three errors of EfficientNetB0 and of ResNet50 lie in this subset, so both models are error-free on the 403 test images that do have a training near-duplicate. On 85 images a single error is worth 1.18 percentage points, so this subset cannot separate the four leaders either, and it is a proxy rather than a substitute for retraining on a deduplicated pool, since the training split still contains the duplicates. Accuracy at other tolerances is given in Supplementary Table S12 and Figure S21.

InceptionV3 followed at 98.77% accuracy with an AUC of 0.9986. The three transformer and hybrid models diverged sharply: CoAtNet-0, which combines convolution with attention, matched the best CNNs, whereas the two pure transformers trailed, with Swin-Base at 95.90% and ViT-B/16 at 94.88%. MobileNetV3-Large reached only 90.78% accuracy despite an AUC of 0.9981, and this gap between ranking quality and accuracy is diagnostic of a badly placed decision threshold rather than poor discrimination: its sensitivity was 81.85% against a specificity of 100.00%, its expected calibration error of 0.0685 was the worst in the study, and its optimal Youden threshold of 0.00044 lies some three orders of magnitude below the 0.5 decision point actually used, so that 45 tumours are missed purely through threshold placement and recalibration alone would recover most of the loss. Its validation loss of 0.3995 against a training accuracy of 99.90% at the final epoch indicates substantial overfitting. VGG16 performed worst at 88.32% accuracy with an AUC of 0.9676, consistent with the expectation that a 134-million-parameter architecture without residual or dense connectivity is poorly suited to a dataset of this size. The weaker models lose more on the leakage-free subset, InceptionV3 falling to 95.29%, ViT-B/16 to 91.76%, Swin-Base to 89.41%, VGG16 to 82.35%, and MobileNetV3-Large to 80.00%, so the gap between the leading group and the rest widens once memorised near-duplicates are excluded. Training and validation curves, receiver operating characteristic and precision–recall curves, and reliability diagrams for all nine binary models are given in Supplementary Figures S7, S9, and S11, and the per-class metric summary, representative predictions, and validation and row-normalised confusion panels of EfficientNetB0 in Supplementary Figures S12, S13, and S14.

Table 7: Binary detection results on the held-out test split (488 images). Sensitivity is recall of the tumour-positive class and specificity is recall of the tumour-negative class. Parameter counts are measured from the instantiated models. "Acc. LF" is accuracy on the 85 leakage-free test images, those with no training image within Hamming distance 5 (Section 3.3). Rows are ordered by accuracy.

Model	Acc. (%)	Sens. (%)	Spec. (%)	F1 (%)	AUC	Test loss	Params (M)	Acc. LF (%)
CoAtNet-0	99.39	100.00	98.75	99.40	0.9999	0.0253	26.67	98.82
DenseNet121	99.39	98.79	100.00	99.39	1.0000	0.0189	6.95	97.65
EfficientNetB0	99.39	100.00	98.75	99.40	0.9998	0.0233	4.01	96.47
ResNet50	99.39	98.79	100.00	99.39	1.0000	0.0217	23.51	96.47
InceptionV3	98.77	97.98	99.58	98.78	0.9986	0.0419	24.35	95.29
Swin-Base	95.90	93.55	98.33	95.87	0.9961	0.1117	86.74	89.41
ViT-B/16	94.88	96.37	93.33	95.03	0.9862	0.1525	85.80	91.76
MobileNetV3-Large	90.78	81.85	100.00	90.02	0.9981	0.3698	4.20	80.00
VGG16	88.32	86.29	90.42	88.25	0.9676	0.2448	134.26	82.35



(a) EfficientNetB0.

(b) ResNet50.

Figure 5: Test-split confusion matrices for **EfficientNetB0** and **ResNet50** on the binary detection task, both at 99.39% accuracy. The three EfficientNetB0 errors are false positives on tumour-negative images, whereas the three ResNet50 errors are missed tumours. Validation and row-normalised panels are given in Supplementary Figure S14.

4.2. Multiclass Classification Results

Test-split results for classification into glioma, meningioma, and pituitary subtypes are reported in Table 8, and the per-class breakdown in Supplementary Table S5. Swin-Base achieved the highest accuracy at 98.58%, corresponding to 10 misclassified images out of 706, with a balanced accuracy of 98.42%, a macro F1 score of 98.47%, and the lowest test loss in the study at 0.0410. CoAtNet-0 followed at 98.44% and InceptionV3 at 98.30%, the latter achieving the highest macro one-versus-rest AUC at 0.9994. EfficientNetB0 and ResNet50 tied at 97.59% accuracy, which is notable given that EfficientNetB0 uses 4.01 million parameters against 23.51 million for ResNet50, a difference of almost six times for identical accuracy. ViT-B/16 was again the weakest model by a substantial margin at 90.51% accuracy and a macro AUC of 0.9759. Meningioma is consistently the hardest class: Swin-Base, CoAtNet-0, and EfficientNetB0 all reached a meningioma sensitivity of 95.92%, well below their glioma and pituitary sensitivities, and meningioma carries the lowest F1 score for five of the six models. This is clinically plausible, since meningiomas arise from the meningeal layers and can present with less distinctive intensity and shape signatures on a single two-dimensional slice than the typically well-localised pituitary lesions, and meningioma is also the smallest class, with 913 training images against 1,400 for glioma. The test-split confusion matrices confirm the pattern: of the 10 Swin-Base errors in Figure 6, 8 are meningioma cases assigned to another class and no pituitary case is misclassified, while EfficientNetB0 in Figure 7 makes 17 errors on the same 706 images, seven more for a twenty-onefold reduction in parameter count. Because this pool was deduplicated before splitting, the multiclass figures are robust to the leakage audit: on the 703 test images without an NCC duplicate in training, no model's accuracy changes by more than 0.04 percentage points. Even on the 265 test images with no training image within Hamming distance 8, a stricter criterion than near-duplication, accuracy stays between 96.23% and 98.11% for the five stronger models and at 90.19% for ViT-B/16, with InceptionV3 at 98.11% and Swin-Base at 97.74% (Supplementary Table S12).

The contrast between Swin-Base and ViT-B/16 is the clearest architectural signal in these results. Both are transformer backbones of almost identical size, 86.75 million and 85.80 million parameters respectively, yet they differ by just over eight percentage points in accuracy. Swin-Base imposes a hierarchical, locally windowed attention structure that reintroduces a spatial inductive bias, whereas ViT-B/16 treats the image as a flat sequence of patches and must learn that structure from data; with only 3,293 training images there is insufficient data to learn it, and the pure transformer underperforms. CoAtNet-0 supports the same reading from the other direction, reaching 98.44% by combining convolutional stages with attention stages at under a third of the parameters of either pure transformer. The conclusion is not that attention is unhelpful for this task but that, at this data scale, attention needs to be paired with a convolutional prior. This reading should be qualified by the interpretability evidence in Section 4.4, which indicates that part of the Swin-Base advantage may rest on non-anatomical image cues; the architectural argument and the shortcut explanation are not mutually exclusive, and these experiments cannot separate them. Training curves for all six multiclass models are given in Supplementary Figure S8, their one-versus-rest receiver operating characteristic and precision–recall curves in Supplementary Figure S10 and reliability diagrams in Supplementary Figure S11, and the per-class metric summary, representative predictions, and validation and row-normalised confusion panels of EfficientNetB0 in Supplementary Figures S12, S13, and S14.

Table 8: Multiclass classification results on the held-out test split (706 images). Precision and F1 are macro-averaged across the three tumour classes, and AUC is the macro-averaged one-versus-rest value. Rows are ordered by accuracy.

Model	Acc. (%)	Bal. acc. (%)	Prec. (%)	F1 (%)	AUC	Test loss	Params (M)
Swin-Base	98.58	98.42	98.54	98.47	0.9993	0.0410	86.75
CoAtNet-0	98.44	98.21	98.50	98.35	0.9989	0.0521	26.67
InceptionV3	98.30	98.29	98.17	98.22	0.9994	0.0471	25.12
EfficientNetB0	97.59	97.50	97.41	97.45	0.9988	0.0588	4.01
ResNet50	97.59	97.57	97.36	97.46	0.9988	0.0593	23.51
ViT-B/16	90.51	90.37	90.31	90.34	0.9759	0.2637	85.80

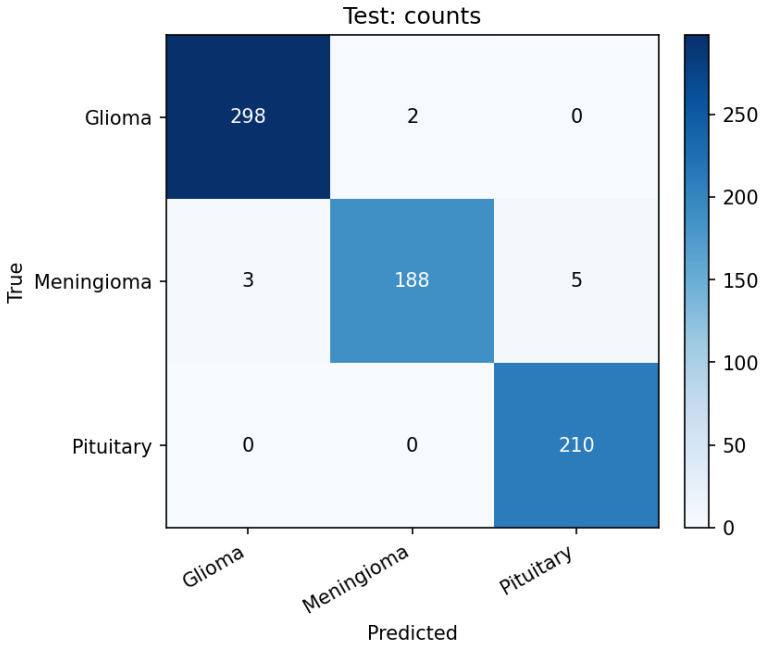


Figure 6: Test-split confusion matrix for **Swin-Base**, the best-performing subtype classifier. Of the 10 errors, 8 involve meningioma; no pituitary case is misclassified. Validation and row-normalised panels are given in Supplementary Figure S14.

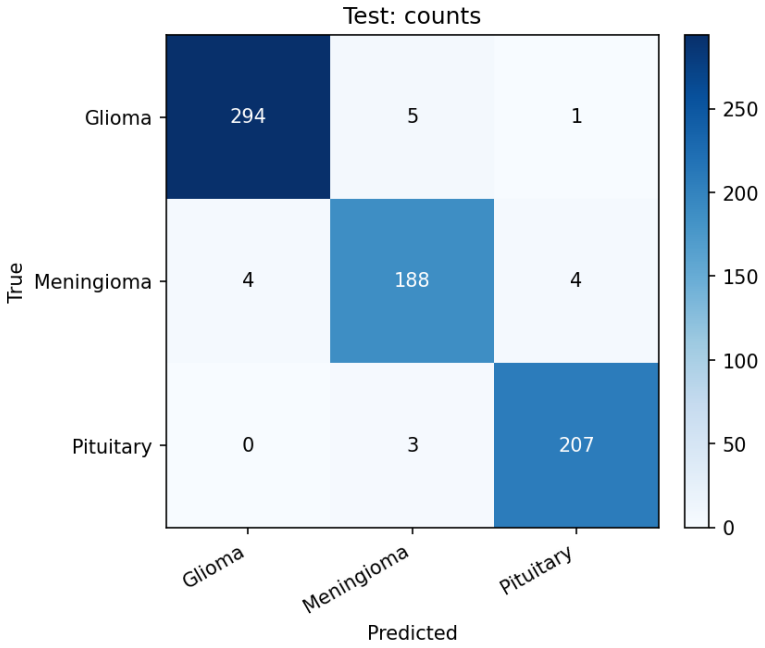


Figure 7: Test-split confusion matrix for **EfficientNetB0** on the multiclass classification task, with 17 errors in total. Most residual confusion occurs between meningioma and the other two classes. Validation and row-normalised panels are given in Supplementary Figure S14.

4.3. Model Efficiency Comparison

Table 9 compares computational cost for both tasks in paired columns, with all measurements taken on the same Tesla T4 hardware; the complete per-task measurements, including batch size and training time, are given in Supplementary Tables S9 and S10. Three distinct efficiency regimes are visible. The lightweight CNNs, EfficientNetB0 and MobileNetV3-Large, require under 0.8 GFLOPs per image and deliver the highest batched throughput, at 1,636 and 2,665 images per second respectively on the binary task. The mid-sized convolutional and hybrid models, DenseNet121, ResNet50, InceptionV3, and CoAtNet-0, occupy an intermediate band of 5.7 to 11.4 GFLOPs. The InceptionV3 figure of 11.42 GFLOPs is not like-for-like with the others, because it is computed at the native InceptionV3 input of 299×299 pixels rather than the 224×224 pixels used by every other model; with 1.78 times as many input pixels, its cost per image is inflated relative to the rest of the band, and its position at the top of that band reflects input resolution as well as architecture. The large transformers and VGG16 sit at 22.5 to 30.9 GFLOPs and deliver the lowest throughput, with Swin-Base managing only 215 images per second on the binary task, roughly one twelfth of MobileNetV3-Large.

The accuracy-per-million-parameters columns make the architectural efficiency difference stark. On the binary task EfficientNetB0 achieves 24.79 percentage points of accuracy per million parameters, MobileNetV3-Large 21.60, and DenseNet121 14.29, while the transformers deliver 1.11 and VGG16 only 0.66; EfficientNetB0 is therefore more than thirty-seven times more parameter-efficient than VGG16 while also being eleven percentage points more accurate, and on the multiclass task it again leads at 24.33 points per million, more than twenty times the figure for either transformer. Wall-clock training time does not track parameter count, because batch size, input resolution, and attention cost all intervene: CoAtNet-0 required 29.1 minutes for the binary task while EfficientNetB0 required 16.3 minutes despite being the smallest model in the study, a consequence of CoAtNet-0 running at batch size 64 against 256 and therefore taking four times as many optimiser steps (Supplementary Table S9). The binary InceptionV3 training time of 4.9 minutes should be disregarded for comparison purposes: as described in Section 3.6, that run executed on a single GPU rather than two, so its wall-clock figure is not measured under the same conditions as the other eight rows. Its parameter count, GFLOPs, latency, and throughput are unaffected, since those were measured separately on one device for every model.

Latency and throughput were re-measured after training so that both could be reported with a variance measure, using randomly initialised copies of each architecture, since compute cost does not depend on the weight values. Single-image latency was measured on one Tesla T4 at batch size one under half-precision autocasting, using a synthetic input at each model's native resolution, over 100 timed forward passes that followed 10 untimed warm-up passes, with the device synchronised before each timing was read. Throughput was measured at batch size 32 over 20 timed passes that followed 3 warm-up passes, and computed per pass. The whole protocol was repeated three times per model, with no other work on the device; Table 9 reports the mean and the within-run standard deviation, and the standard deviation between the three repeat means is at most 0.6 ms for latency and 18 images per second for throughput (Supplementary Figure S22). The original single-session measurements, taken on the same hardware type under PyTorch 2.10.0 rather than 2.11.0, were 0.8 to 7.4 ms slower for every model, and the two EfficientNetB0 measurements, which then differed by 3.1 ms between tasks, now agree within 0.1 ms. Latency differences of a few milliseconds measured in different sessions should therefore not be over-interpreted. With that caveat, single-image latency is lowest for VGG16 at 4.6 ± 0.1 ms, followed by ResNet50 at 6.6 ± 0.2 ms, the fastest of the four leading binary detectors, against 8.9 ± 0.4 ms for EfficientNetB0, because the depth-wise separable convolutions that make EfficientNetB0 parameter-efficient do not map as efficiently onto the GPU at batch size one. Latency also separates the two transformers sharply despite their near-identical size: on the multiclass task Swin-Base requires 27.8 ± 1.2 ms against 7.5 ± 0.4 ms for ViT-B/16, because the shifted-window attention that gives Swin-Base its accuracy advantage also introduces sequential windowing operations that do not parallelise well at batch size one. Swin-Base therefore buys its leading accuracy at nearly four times the per-request latency of ViT-B/16 and about three times that of EfficientNetB0. For deployment, the correct efficiency metric depends on whether the constraint is memory footprint, batch throughput, or single-request latency.

Table 9: Computational efficiency for the binary (Bin.) and multiclass (Multi.) tasks in paired columns. Latency is the mean ± within-run standard deviation of 100 timed single-image forward passes after 10 warm-up passes; "Thr." is throughput in images per second at batch size 32, the mean ± within-run standard deviation of 20 timed passes after 3 warm-up passes; both are pooled over three repeats of the protocol. "Acc./M" is test accuracy in percentage points per million parameters. All inference measurements used one Tesla T4 under half-precision autocasting. A dash marks a model not run on the multiclass task. Batch sizes and training times are given in Supplementary Tables S9 and S10. Rows are ordered by parameter count.

			Latency (ms)		Thr. (img/s)		Acc./M	
Model	Params (M)[a]	GFLOPs	Bin.	Multi.	Bin.	Multi.	Bin.	Multi.
EfficientNetB0	4.01	0.77	8.9 ± 0.4	8.9 ± 0.6	1636 ± 20	1538 ± 34	24.79	24.33
MobileNetV3-Large	4.20	0.43	7.1 ± 0.3	–	2665 ± 59	–	21.60	–
DenseNet121	6.95	5.67	16.7 ± 0.9	–	707 ± 7	–	14.29	–
ResNet50	23.51	8.17	6.6 ± 0.2	6.7 ± 0.3	819 ± 4	771 ± 10	4.23	4.15
InceptionV3	24.35	11.42[b]	12.3 ± 0.4	12.2 ± 0.8	564 ± 3	525 ± 4	4.06	3.91
CoAtNet-0	26.67	8.42	11.6 ± 0.4	11.8 ± 0.4	578 ± 3	516 ± 5	3.73	3.69
ViT-B/16	85.80	22.54	7.4 ± 0.6	7.5 ± 0.4	351 ± 3	329 ± 2	1.11	1.05
Swin-Base	86.74	30.86	24.0 ± 0.5	27.8 ± 1.2	215 ± 1	205 ± 1	1.11	1.14
VGG16	134.26	30.93	4.6 ± 0.1	–	397 ± 2	–	0.66	–

[a]Binary models; the multiclass InceptionV3 and Swin-Base models measure 25.12 and 86.75 million parameters (Section 3.6).

[b]Computed at the native 299 × 299 input; every other model is measured at 224 × 224, so this value is not like-for-like.

Calibration quality, reported in Table 10, adds a dimension that accuracy alone does not capture. The four leading binary models are all well calibrated, with expected calibration errors between 0.0044 and 0.0084, meaning that a predicted confidence of 95% corresponds closely to a 95% empirical success rate. MobileNetV3-Large and VGG16, by contrast, show calibration errors of 0.0685 and 0.0437, an order of magnitude worse. This matters for deployment: a model whose confidence scores are unreliable cannot safely be used to triage cases by confidence or to defer uncertain cases to a human reader, even if its ranking performance measured by AUC appears strong. On the multiclass task all six models are reasonably calibrated, with Swin-Base best at 0.0076 and ViT-B/16 worst at 0.0266.

Table 10: Expected calibration error (ECE), Matthews correlation coefficient (MCC), and Cohen's kappa on the held-out test split for both tasks. Lower ECE indicates better agreement between predicted confidence and observed accuracy.

Task	Model	ECE	MCC	Kappa
Binary	DenseNet121	0.0044	0.9878	0.9877
	EfficientNetB0	0.0049	0.9878	0.9877
	ResNet50	0.0057	0.9878	0.9877
	InceptionV3	0.0078	0.9755	0.9754
	CoAtNet-0	0.0084	0.9878	0.9877
	Swin-Base	0.0235	0.9192	0.9181
	ViT-B/16	0.0323	0.8978	0.8975
	VGG16	0.0437	0.7673	0.7665
	MobileNetV3-Large	0.0685	0.8302	0.8161
Multiclass	Swin-Base	0.0076	0.9784	0.9783
	InceptionV3	0.0077	0.9741	0.9740
	CoAtNet-0	0.0084	0.9762	0.9761
	EfficientNetB0	0.0112	0.9632	0.9632
	ResNet50	0.0116	0.9633	0.9632
	ViT-B/16	0.0266	0.8550	0.8550

4.4. Interpretability Results

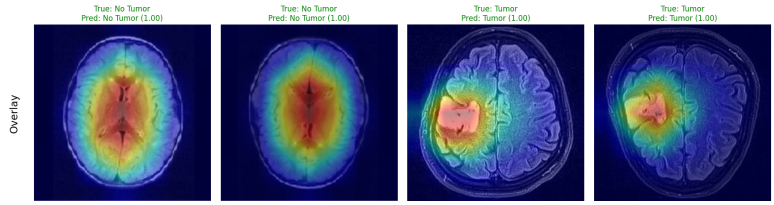
A performance figure alone does not establish that a model is attending to clinically relevant image regions rather than to incidental artefacts such as skull outline, field of view, or scanner-specific intensity patterns. To examine this, gradient-weighted class activation mapping was applied to EfficientNetB0 for both tasks and to Swin-Base for the multiclass task. In every case the gradients of the predicted class score with respect to the target-layer activations were averaged spatially to give one weight per channel, the weighted sum of the activations was passed through a rectified linear unit, upsampled bilinearly to the input resolution, and min–max normalised. For EfficientNetB0 the target layer is the final convolutional stage. The transformer backbones require a further choice that materially changes the resulting map. For Swin-Base the target is the first layer normalisation of the last block of the final stage; its output is already a 7×7 spatial grid in channels-last order, so the only transform applied is a permutation to channels-first order. For ViT-B/16 the target is the first layer normalisation of the last encoder block, whose output is a token sequence; the class token is discarded and the remaining 196 patch tokens are reshaped to a 14×14 grid before the map is computed. No attention-native method such as attention rollout was applied to corroborate the transformer maps. Two caveats therefore apply before any comparison is made. First, the activation maps of convolutional and attention stages are not strictly commensurable, because they are taken from layers of different kinds and spatial resolution, so contrasts between EfficientNetB0 and Swin-Base below are prompts for further investigation rather than measured differences. Second, the transformer maps depend on the target layer chosen, and a different layer could yield a different map.

Figure 8 shows the EfficientNetB0 maps for the most confident correctly classified test cases of both tasks; the full input, activation, and overlay panels are given in Supplementary Figure S15. For correctly classified tumour-positive cases, the activation consistently concentrates on the lesion itself rather than on peripheral anatomy, which supports the interpretation that the model has learned a tumour-relevant representation for this backbone. For correctly classified tumour-negative cases the activation covers essentially the whole brain; this is the expected behaviour for a negative class, since there is no lesion to localise and the decision rests on the absence of a focal abnormality across the brain, and it should not be read as diffuse or unfocused attention. In the multiclass setting the attended regions are broadly consistent with the anatomical expectations for each subtype, with pituitary cases showing focused activation in the sellar region and glioma cases showing more diffuse activation across the affected hemisphere.

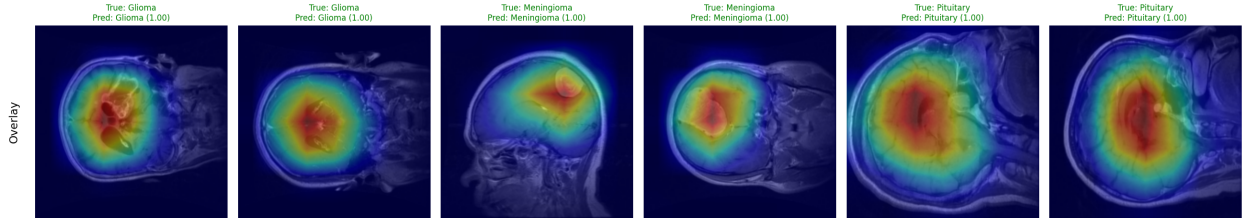
Figure 9 shows the equivalent maps for Swin-Base on the multiclass task, and the comparison is not reassuring. For meningioma the Swin-Base heat maps sit on the visible lesion, but for glioma and pituitary they largely highlight image corners and borders lying outside the brain entirely. Since Swin-Base is the most accurate subtype classifier in this study, this is a material finding: it suggests that part of its accuracy may derive from background or acquisition cues that correlate with class rather than from tumour morphology. Such cues are plausible here because the two source collections differ in native image size and framing, as Tables 2 and 3 show, and because class proportions differ between them, so border artefacts could carry genuine label information without carrying any clinical meaning. This observation remains qualitative and drawn from a single figure: the Swin-Base checkpoint is not among the released artefacts, so the measurements below could not be applied to it, and no measurement against annotated masks was performed. It nevertheless tempers the headline ranking, since a model that attains 98.58% accuracy partly by reading image borders would not retain that accuracy on data acquired under a different protocol, and it illustrates why interpretability evidence belongs alongside accuracy rather than after it.

For EfficientNetB0, whose checkpoints are released, attention was quantified after training; reloaded, both checkpoints reproduce the stored test predictions exactly. A head mask was computed for every test image by Otsu thresholding, a small morphological closing, selection of the largest connected component, and hole filling. On these two-dimensional slices the mask covers brain and skull, and on some sagittal slices the neck, so energy outside it falls on background, borders, and framing (Supplementary Figure S23). The share of Grad-CAM energy outside the mask, as mean ± standard deviation over test images, is 0.068 ± 0.118 for tumour-negative and 0.077 ± 0.118 for tumour-positive binary images, and 0.040 ± 0.020 for glioma, 0.090 ± 0.044 for meningioma, and 0.092 ± 0.110 for pituitary, against 0.39 to 0.53 for a uniform map over the same images (Supplementary Figure S24); restricting to correct predictions changes these values by at most 0.002. An inference-only masking test agrees (Supplementary Figure S25). With the head region set to zero, binary accuracy falls from 99.39% to 55.74%, 4.9 points above the majority-class rate of 50.82%, and multiclass accuracy from 97.59% to 36.97%, below the majority-class rate of 42.49%; with the background set to zero instead, accuracy is largely retained, at 97.75% and 96.60%. EfficientNetB0 therefore relies predominantly on the head region, with at most a small background signal in the binary task. A fall to chance on background-only images does not rule out a shortcut, however, because such images lie far from the training distribution; only retraining on background-only images, which was not performed, can settle that question.

Maps for misclassified cases are given in Supplementary Figure S16. The selection rule matters for their interpretation: these panels show the *most confident* errors, up to six per model, ranked by the probability assigned to the wrong class, and are therefore the failures in which the model was most assured, not representative or borderline ones. For EfficientNetB0 the binary panel exhausts the available errors, since only three test images were misclassified, whereas the multiclass panel shows six of seventeen. In these high-confidence failures the activation is not noticeably less focused than in the correct cases. All of these observations are qualitative rather than a quantitative interpretability evaluation. No assessment against radiologist-annotated tumour masks was carried out, because the datasets used provide class labels only and contain no segmentation masks against which the attention maps could be scored. Quantifying attention overlap against expert annotation is identified in Section 5 as necessary future work.



(a) Binary detection: two tumour-negative and two tumour-positive cases.



(b) Multiclass classification: two cases each of glioma, meningioma, and pituitary.

Figure 8: Gradient-weighted class activation maps for **EfficientNetB0**, overlaid on the most confident correctly classified test cases of each class. Activation concentrates on the lesion for tumour-positive cases and covers the whole brain for tumour-negative cases, where there is no lesion to localise. The full input, activation, and overlay panels are given in Supplementary Figure S15.

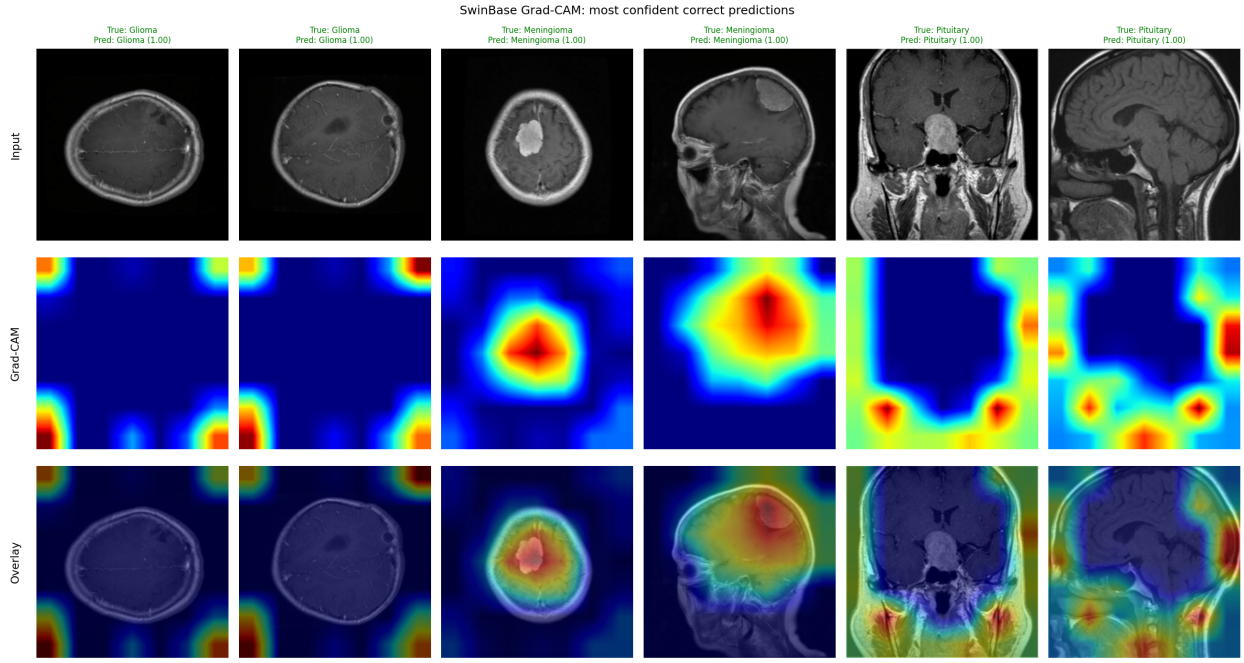


Figure 9: Gradient-weighted class activation maps for **Swin-Base** on correctly classified multiclass test cases, showing the two most confident correct predictions per class. Activation sits on the lesion for meningioma but falls largely on image corners and borders outside the brain for glioma and pituitary, suggesting partial reliance on non-anatomical cues.

4.5. Comparison with Existing Literature

Table 11 compares the present results with previously published studies using the same architectures. The ResNet50 and InceptionV3 results obtained here are consistent with, or exceed, the published benchmarks. Two discrepancies deserve comment. The VGG16 result of 88.32% falls well short of the 98.69% reported by Khan et al. [16]; this is attributable to differences in preprocessing, augmentation strategy, and train-test partitioning, and in particular to the fact that the present protocol fixes a single shared set of hyperparameters across all nine models rather than tuning each one individually, a choice that penalises architectures such as VGG16 that are sensitive to learning rate. The DenseNet121 result of 99.39% is marginally below the 99.51% reported by Noreen et al. [17], a difference of less than one image on a test set of this size and therefore not meaningful.

More generally, direct comparison with published figures should be treated with caution, for three reasons. First, most of the studies in Table 11 evaluate on the 3,064-image Cheng collection without deduplication, whereas the multiclass results reported here are computed after removing 19.3% of the images as near-duplicates; deduplication removes exactly the images most likely to have inflated a reported score, so a slightly lower number obtained on a deduplicated split is not evidence of a weaker model. Conversely, the binary figures reported here are themselves inflated by train-test leakage, and on the 85 leakage-free test images the ResNet50, DenseNet121, and InceptionV3 figures in the final column fall to 96.47%, 97.65%, and 95.29%. Second, as noted in Section 3.6, the multiclass pool here merges the original training and testing folders of the Bhuvaji dataset before drawing fresh stratified splits, so the test set used in this work is not the test set used by studies that adopt that dataset's official partition. Third, the models compared here share a single hyperparameter configuration rather than being individually tuned. The comparison is therefore included for orientation rather than as a like-for-like benchmark, and the absolute figures in the final column should not be cited as head-to-head wins or losses against the referenced studies.

Table 11: Comparison of the present results with previously published studies. Results obtained in this work are measured on the held-out test split.

Study	Model	Dataset size	Task	Reported accuracy	This work
Talo et al. (2019) [13]	ResNet50	253	Binary	95.23%	99.39%
Khan et al. (2020) [16]	VGG16	3,064	Binary	98.69%	88.32%
Noreen et al. (2020) [17]	DenseNet121	3,064	Binary	99.51%	99.39%
Rehman et al. (2020) [18]	InceptionV3	3,064	Binary	99.34%	98.77%
Deepak and Ameer (2019) [12]	InceptionV3	3,064	Multiclass	97.50%	98.30%

Several architectural innovations contribute to improved performance in medical imaging tasks. Residual learning in ResNet enables the training of deeper networks that capture hierarchical representations from low-level edges to high-level semantic features. Dense connections maximise information flow and feature reuse with comparatively few parameters, which explains why DenseNet121 matches the accuracy of ResNet50 with less than a third of its parameters. Inception modules capture tumours of varying size through parallel convolutional paths of different kernel sizes. Attention mechanisms in CoAtNet-0 and Swin-Base model long-range dependencies that are relevant for relating a tumour region to surrounding healthy tissue. Neural architecture search in EfficientNetB0 and MobileNetV3-Large optimises architectural decisions beyond human intuition, though the MobileNetV3-Large result shows that architectural efficiency does not by itself guarantee stable optimisation under a fixed shared training budget.

The clearest cross-cutting finding is what may be termed the transformer paradox: pure transformers underperformed while hybrid convolution-attention models excelled. ViT-B/16 was the weakest model on both tasks despite being among the largest, whereas CoAtNet-0 was among the strongest on both. With training sets of 2,277 and 3,293 images, there is insufficient data for a pure transformer to learn the spatial inductive biases that convolutional architectures encode by construction. The practical implication for medical imaging at this data scale is that attention should be adopted in hybrid form rather than in isolation.

Regarding model selection, the appropriate choice depends on the deployment constraint. Where accuracy is the primary consideration and computational resources are not limiting, Swin-Base is the strongest subtype classifier at 98.58% accuracy, and CoAtNet-0 offers almost the same accuracy at 98.44% for less than a third of the parameters and roughly two and a half times the throughput. Where deployment is on constrained hardware, EfficientNetB0 is the clear recommendation: it reaches 99.39% detection accuracy and 97.59% subtype accuracy with 4.01 million parameters, 0.77 GFLOPs per image, and well-calibrated confidence scores. Where single-request latency dominates, ResNet50 provides the fastest single-image inference among the leading detectors at 6.6 ± 0.2 ms while retaining 99.39% detection accuracy, with only the far less accurate VGG16 faster, and it remains a reasonable default baseline for further research. No single model dominates on every axis, and the evidence presented here is intended to support a selection made against explicit deployment requirements.

The misclassification panels of Section 4.4 carry a further implication for deployment. They show the most confident errors, and in those errors the activation maps are as plausible-looking as in correct predictions. Confidently wrong predictions with plausible-looking attention maps cannot be filtered by confidence thresholding, because the confidence is high, or by visual review of the map, because the map looks reasonable. This is a direct argument against deploying Grad-CAM as a clinical safety check: an explanation map can accompany a prediction, but it cannot be relied on to flag the predictions that are wrong.

5. Conclusion

This study benchmarked nine ImageNet-pretrained architectures for brain tumour detection, and six of them for subtype classification, under matched data, preprocessing, and training schedule, with perceptual-hash deduplication removing 19.3% of the multiclass images before splitting. On binary detection, CoAtNet-0, DenseNet121, EfficientNetB0, and ResNet50 each reached 99.39% test accuracy and are statistically indistinguishable, although the binary pool was not deduplicated, 82.6% of its test images have a training near-duplicate, and on the 85 test images without one their accuracy is 96.47% to 98.82%, while MobileNetV3-Large at 90.78% and VGG16 at 88.32% proved inadequate. On subtype classification, Swin-Base led at 98.58%, followed by CoAtNet-0 at 98.44% and InceptionV3 at 98.30%, with EfficientNetB0 and ResNet50 at 97.59% and ViT-B/16 trailing at 90.51%.

Three findings carry beyond the leaderboard. First, the hybrid CoAtNet-0 consistently outperformed the pure transformer ViT-B/16 of three times its size, indicating that at this data scale attention should be paired with a convolutional prior. Second, parameter efficiency and accuracy are not in tension: EfficientNetB0 matched or approached far larger models with 4.01 million parameters and 0.77 GFLOPs per image. Third, headline accuracy conceals clinically material differences: models with identical accuracy differed in whether their errors were missed tumours or false alarms, MobileNetV3-Large combined a strong AUC with poor calibration, and the Grad-CAM evidence indicates that the most accurate subtype model attends partly to regions outside the brain, so accuracy alone does not establish that pathology has been learned.

The principal limitations are as follows. The datasets are public two-dimensional slice collections of unknown MRI sequence composition, the binary negative class is tumour-negative rather than certified normal, and no external validation was performed. Each model was trained once for ten epochs with a single shared configuration, and batch size varied from 32 to 256, so the optimisation budget is uncontrolled and the leading models cannot be separated statistically. Deduplication was applied before training to the multiclass track only, the binary models were not retrained on the deduplicated binary pool, and splits were drawn at slice rather than patient level. Swin-Base was sourced from different implementations in the two tasks, and the interpretability evidence for Swin-Base is qualitative, since attention outside the head was measured only for EfficientNetB0.

Future work should retrain the binary models on the deduplicated binary pool and the multiclass models on the label-aware pool; split by patient where identifiers are available; equalise the optimisation budget by fixing the number of gradient updates; adopt repeated runs with cross-validation; extend the outside-head attention measurement to the remaining models and test for shortcut learning by retraining on background-only images; and validate externally on multi-sequence and three-dimensional institutional data before any clinical claim is made. The architectures examined here are accurate enough to warrant investigation as second-reader aids rather than autonomous diagnostics, and the accompanying code and per-model artefacts are released so that both the results and their caveats can be reproduced and extended.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this study are publicly available and are cited in Section 3.2. The complete code, trained model checkpoints, and per-model evaluation artefacts are available at the repository given in Appendix A.

Supplementary material

Supplementary material associated with this article is provided in a separate file. It contains the per-class detection and classification metrics (Supplementary Tables S4 and S5), the training and validation metrics at the checkpointed epoch (Supplementary Tables S6 and S7), the complete efficiency measurements for each task (Supplementary Tables S9 and S10), the per-source deduplication outcome and the test accuracy on leakage-free subsets (Supplementary Tables S11 and S12), and Supplementary Figures S1–S25, covering dataset characterisation, the deduplication and leakage audit, augmentation, training curves, receiver operating characteristic, precision–recall and reliability curves for every model, per-class and prediction summaries, full confusion panels, timing variance, the complete gradient-weighted class activation maps for correct and misclassified cases, and the outside-head attention and masking measurements.

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Appendix A. Code Repository

The complete code for all experiments, including data preprocessing, perceptual hash deduplication, model training, evaluation, and figure generation, together with the trained checkpoints and the full per-model metric artefacts, is available at https://github.com/parthmital/NeuroScan-AI.